

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-3;

BASIC DATA; KEY CHALLENGES;

CROPPING PATTERN

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2. AGRICULTURE'S SIGNIFICANCE FOR INDIA (ESI 2023-24)

- **GDP Share:** 18.2% of country's GDP
 - » **Note:** At the time of Independence, Agriculture contributed to more than 50% of the GDP and employed 2/3rd of the population.
- **Livelihood support** to 42.3% of country population.
- **Growth** (over past five years): 4.18% (shows buoyancy of the sector).
- **Growth** (FY24): 1.4% (lower than 4.7% in FY23)
 - » **Reasons for decline:** Drop in food grains production due to delayed monsoon caused by El-Nino.
- **Share of livestock and fishery sector has been growing:**

	FY23	FY15
Crops	55.28%	61.75%
Livestock	30.23%	24.38%
Fisheries	7.25%	4.44%
Forestry	7.24%	9%

- **Production:** India is the largest producer of milk, pulse and spices. It is also the second largest producer of rice, wheat, cotton, among other crops.

3. AGRICULTURE CENSUS 2015-16

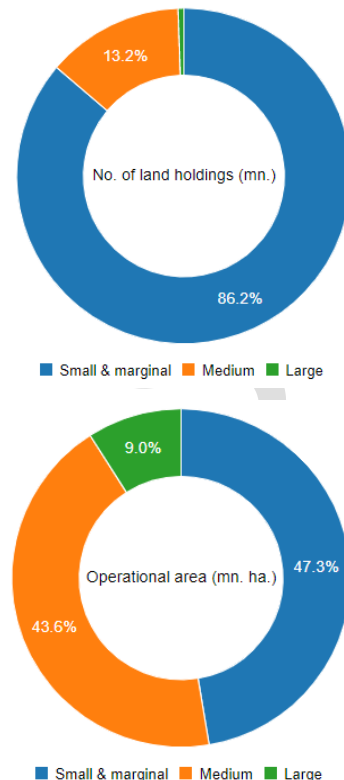
- **Intro**
 - » Department of Agriculture, Cooperation and Farmers Welfare, MoA&FW conducts a **quinquennial Agriculture Census** (every five years) in the country to collect **key information about the structure and agriculture holdings in the country** and **monitor changes that take place over time**.
 - » **2015-16** census was the **10th Agri census** conducted in the country. (the first census was done in 1970-71)
 - **Note:** As of Jan 2024, 11th Agri-Census is going on. It was launched in July 2022.
 - » In the census, the unit of enumeration is defined by the term '**Operational Holding**' and it corresponds to the person who actually cultivates the land rather than its ownership. The concepts followed are broadly in conformity with the World Census of Agriculture, FAO of the UN.

- **Term to understand: Size Classes**

S.No.	Group	S.No.	Classes (in ha.)
I.	Marginal	1.	Below 0.5 ha.
		2.	0.5 – 1.0 ha.
II.	Small	3.	1.0 – 2.0 ha.
III.	Semi-medium	4.	2.0 – 3.0 ha.
		5.	3.0 – 4.0 ha.
IV.	Medium	6.	4.0 – 5.0 ha.
		7.	5.0 – 7.5 ha.
		8.	7.5 – 10.0 ha.
V.	Large	9.	10.0 – 20.0 ha.
		10.	20.0 and above.

- Key Highlights

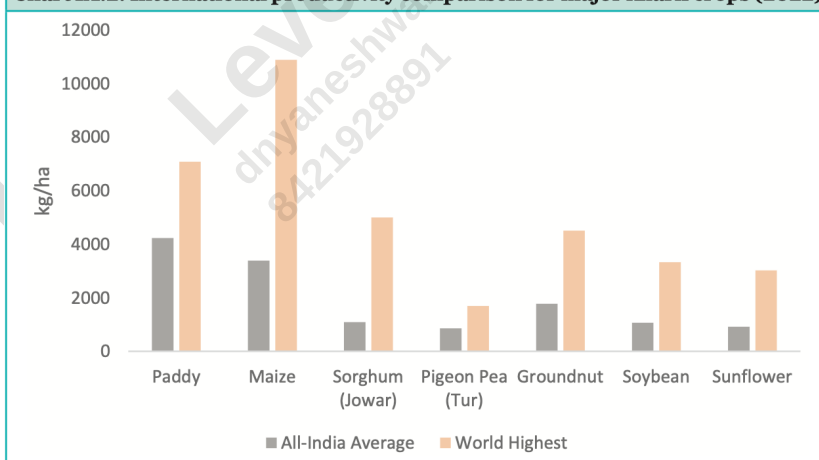
- **Decline in total operated area by 1.53%:** From 159.59 million hectares to 157.14 million hectares.
- **State wise total operated area is highest in: Rajasthan,** followed by Maharashtra, UP, MP and Karnataka.
- **Small and marginal landholdings (<2 hectare area)** constituted **86.21%** of the total landholding, an **increase of 1.2%** points compared to 2010-11.
 - Farmers holding 10 hectares and more account for just **0.57%**.
 - Semi-Medium and Medium: 13.2%
- It is noteworthy that **small, marginal and medium landholdings** constitute the lion's share of operated area - **large landholding** account for only 9% of the total operated area.



4. KEY PROBLEMS FACED BY AGRICULTURE SECTOR

- **Small landholdings: Fragmentation** -> Hinders investment and increase the input cost
- **Insufficient access to quality input**
- **High dependency on Monsoon; Insufficient and inefficient irrigation system**
 - Fluctuation of monsoon leads to fluctuation of agri-output
 - Flood irrigation which is very inefficient is very common, whereas the number of farmers using drip or sprinkler irrigation is very less.
- **Low Productivity:** The crop yield in the country is much lower than the other major producers.

Chart IX.2: International productivity comparison for major kharif crops (2022)



Source: Price Policy Report for Kharif Crops 2024-25

- **Poor Agri-Marketing and Agri-marketing infrastructure** -> It is leading to high post-harvest losses.
- **Disguised unemployment: Large number of workforce employed in agriculture**
 - Low level of mechanization;

- High input cost
- **Volatility in income for farmers:** Very few farmers are able to get benefit of MSPs
- **Poor Land leasing and contract farming penetration**
- **Sustainability Concerns:**
 - Water Table decline;
 - Excessive fertilizer use -> Nitrogen Pollution -> Land degradation
 - Global warming: Methane Emission; stubble burning; Nitrous Oxide Emissions etc.

1) SEVERAL INTERVENTIONS BY GOVERNMENT

1. Recommendations of the **Doubling Farmers Income (DFI) Report, 2016**, which identifies strategies to improve crop and livestock productivity, enhancing cropping intensity, diversify to high value agriculture and provide remunerative prices on farmers' produce.
 - The 2018 decision to fix MSP at 1.5 times the all India weighted average cost of production was a step towards providing remunerative prices to farmers.
2. **Income Support** through PM-KISAN (Pradhan Mantri Kisan Samman Nidhi), which gives farmers a direct financial benefit of Rs 6,000 per year.
3. **Steps to enhance efficiency** - PMKSY - PDMC: a micro irrigation scheme; Soil Health Card initiative
4. **Steps to enhance sustainability** of agriculture sector:
 - National Mission on Sustainable Agriculture (NMSA), including the use of alternative and organic fertilizers.
5. **Marketing Reforms:** e-National Agriculture Market (e-NAM) facilitates the adoption of smart agriculture technologies, with the later allowing better price discovery.

5. AGRICULTURE PRODUCTION IN INDIA

- As per the ESI 2023-24, in 2022-23, the food grain production in India hit an all time high of 329.7 million tonnes, and oilseeds production reached 41.4 million tonnes.
 - » **Food Grains:** 329.68 million tonnes
 - **Rice:** 135.7 million tonnes
 - **Wheat:** 110.5 million tonnes
 - **Nutri / Coarse Cereal:** 57.3 million tonnes
 - **Maize:** 38 million tonnes
 - **Millet/ Shree Anna:** 17.3 millions tonnes
 - **Pulse:** 26.055 million tonnes
 - **Gram:** 12.27 million tonnes
 - **Tur:** 3.3 million tonnes
 - » **Oil Seeds:** 41.3 million tonnes
 - **Soyabean:** 14.9 million tonnes
 - **Rapeseed and Mustard:** 12.6 million tonnes
 - **Groundnut:** 10.29 million tonnes
 - » **Sugarcane:** 490.3 million tonnes
 - » **Cotton:** 336 lakh bales (of 170 Kg each)
 - » **Jute & Mesta:** 93.92 lakh bales (of 180 Kg each)

- In 2023-24, food grain production is slightly lower at **328.8 million tonnes**, primarily because of poor and delayed monsoon.
 - » **Rice:** 136.7 million tonnes (increased of 9 lakh tonnes)
 - » **Wheat:** 112 million tonnes (increased by 2.3 million tonnes)
 - » **Nutri/ Coarse Cereals:** 54.7 million tonnes
 - **Millets/Shree Anna:** 174.08 lakh tonnes (Marginal increase)

6. CROPPING PATTERN

- Cropping pattern indicates the time and spatial distribution of various type of crops being grown in the country.
- **Factors determining Cropping system**
 - » **Agro-Ecological Setting** four nourishment and appropriateness of a crop
 - Soil and Climatic Parameters
 - » **Socio-Economic factors** like land size, financial resources etc.
 - » **Infrastructure Facilities** including irrigation facilities, storage, trade and marketing, post harvest handling, processing etc.
 - » **Technological development** including enhanced varieties, cultural requirements, mechanization, plant protection etc.
 - » **Potential productivity and financial benefits**

7. CROP DIVERSIFICATION

1) QUESTION1:

- Discuss the key advantages of crop diversification in agriculture. What are the factors hindering crop diversification in the country?
- **Introduction**
 - » Crops diversification refers to **shift in cropping pattern from one or a few crops to other crops which are more profitable, sustainable or less resource intensive**.
- **Advantages Of Crop Diversification:**
 - » **Crop Diversification promotes sustainability:**
 - **Improved Soil Health:** Different crops have varied root structures, nutrient requirements, and residue composition. It reduces the risk of nutrient imbalances or deficiencies.
 - **Reduce use of water and chemical fertilizers** (by diversifying to millets/pulse from rice/wheat)
 - **Planting diverse crops disrupts pest, weed and disease cycle.**
 - For e.g. Certain crops have allelopathic properties or growth habits that suppress weeds. By including such crops in a diversified cropping system, weed growth can be effectively controlled without heavy reliance on herbicide.
 - **Fighting the challenges of Monoculture:** For e.g., states like Punjab and Haryana are seeing, declining productivity, lower fertility, water table decline etc.
 - **Supports biodiversity:** By providing habitats for beneficial insects, birds, and pollinators.

- **Resilience to climate change** as different crops will have varying level of tolerance to changing climatic conditions.
 - » **Food Security and Nutrition** - Crop diversification will promote healthier diets and reduce risk of nutritional deficiency.
 - » **Improves the availability of fodder** for livestock animals.
 - » **Risk Mitigation**: It reduces vulnerability of farmers to adverse weather conditions, pests, diseases, or market fluctuations affecting a particular crop.
 - » **Increased income for farmers** - By diversifying to high value horticulture crops.
- **Steps taken to promote diversification**
- » **Crops Diversification Program** (a sub scheme under RKVY) is being implemented by government in original green revolution states viz. Punjab, Haryana, and in Western Uttar Pradesh to diversify paddy areas towards less water requiring crops like oil seeds, pulses, coarse cereals, agro-forestry, and **shifting of tobacco farmers to alternative cropping system** in tobacco growing states viz. Andhra Pradesh, Bihar, Gujarat, Karnataka, Maharashtra, Odisha, TN, Telangana, Uttar Pradesh and West Bengal wef from 2015-16.
 - Under CDP, assistance is provided to states for conducting cluster demonstration on alternate crops, promotion of water saving technologies, distribution of farm machineries, setting up of value addition facilities, awareness through training etc.
 - » **Crop Diversification through Price Policy** -> Increasing MSP for crops which need to be promoted.
- **Status in India**
- » India has tremendous potential for crop diversification: .
 - » **The Index of Crop Diversification** (used by ESI 2017-18) analysis shows that for India as a whole the crop diversification scenario appears to be stable throughout the period.
- **Factors behind decreasing crop diversification in some states**
- » **Minimum Support Price** mechanism distorts the agri-production towards wheat and rice as they have consistently seen MSP increase and procurement.
 - » **Lack of awareness among farmers** about possibility of other crops creates situation of monoculture.
 - » **Limited Input Availability**: Diversification is constrained by availability of quality seeds, planting materials, and other input required for diversification.
 - » **Risk Perception and Market Volatility**: Farmers often perceive diversification as a risky endeavor, particularly when they are unfamiliar with new crop types or have limited knowledge about market demand and pricing fluctuations.
- **Way Forward**
- » **Rationalize MSP** -> Controlling MSP on wheat/rice; Increasing it on pulses/millets etc.
 - » **Soil Health cards and other such initiatives** which inform farmer's about different crops which are suitable for a particular soil type needs to be further strengthened.
 - » **Enhancing access to inputs and strengthening extension services**.
 - » Initiatives like mechanization, micro-irrigation etc. can contribute to crop diversification and needs to be promoted.

- » **Using emerging technologies** like Precision Agriculture, Crop Modeling & Data Analytics, Genetic Engineering etc. to identify and develop crops which would promote diversification.
- » **Promoting awareness** about the benefits and opportunities of crop diversification.
 - NGOs promoting diversification should be provided with governmental support.
- **Conclusion**
 - » Policymakers, agricultural institutions, and market stakeholders need to collaborate to create an enabling environment that supports and incentivizes crop diversification in agriculture.

2) QUES-2:

- What are the present challenges before crop diversification? How the emerging technologies provide an opportunity for crop diversification? [Mains 2021, 15 marks, 250 words]
- **Intro:**
- **Challenges:**
- **Emerging Technologies in agriculture** present significant opportunities for crop diversification:
 - i. **Precision Agriculture:**
 - **IoT and Sensor Technologies** facilitate real-time monitoring of various parameters, such as soil moisture, temperature etc. This allows farmers to optimize crop management practices and make informed decision regarding crop diversification.
 - Technologies like remote sensing, drones, and satellite imagery, allow farmers to assess field variability and make informed decisions about crop selection and management.
 - ii. **Crop Modelling and Data Analytics:** These tools enable farmers to analyze historical weather patterns, soil data, and crop performance data. By using predictive models and data-driven insights, farmers can identify crops that are best suited for specific environmental conditions, thereby promoting crop diversification.
 - iii. **Genetic Engineering and Biotechnology:** They offer opportunities to develop new crop varieties with enhanced traits, such as disease resistance, drought tolerance, and improved nutritional content.
 - iv. **Indoor and Vertical Farming:** These allow for the cultivation of a variety of crops in controlled environments. By providing optimal conditions such as light, temperature, and nutrient supply, these systems offer the flexibility to grow diverse crops regardless of geographic or climatic limitations.
 - v. **Hydroponics and Aeroponics:** They provide an alternative to traditional soil-based agriculture. These soil-less cultivation methods allow for the precise control of nutrient delivery to plants, enabling the production of a wide range of crops in a limited space. This promotes diversification and year-round cultivation.
 - vi. **Mobile Applications and Farm Management Systems:** These can be used to increase awareness and provide information about market prices, best practices etc. All this can nudge farmers towards diversification.
- **Conclusion:**

- » By leveraging these emerging technologies, farmers can overcome traditional limitations, make data-driven decisions, and explore diverse crop options that were previously unviable or challenging

8. RICE WHEAT CROPPING SYSTEM: BOON TO BANE

1) QUES-1:

- What are the major factors responsible for making rice-wheat system a success? In spite of this success how has this system become bane in India [Main 2020, 15 marks, 250 words]
 - **Introduction:**
 - » The Rice-wheat cropping system is a traditional agricultural practice in South Asia which involves growing rice in wet season (Kharif) and wheat in the dry season (Rabi Season). It is the largest agricultural production cropping system in the world and is prominent in north-western India.
 - **The RWCS** has been highly successful in increasing agri-productivity and farmers income over the last many decades. The key factors behind it include:
 - » **Success of Green Revolution:**
 - Availability of high yielding variety of seeds which are responsive to irrigation
 - Good accessibility of machinery for sowing and harvesting operations
 - » **Favorable agro-climatic conditions:** Rice requires hot and humid conditions, whereas wheat thrives in cooler temperatures and dry conditions. The availability of such complementary condition in India's Monsoon kind of climate allows for the successfully rotation of these crops.
 - » **Synergistic Nutrient Management:** Rice and wheat have different nutrient requirements. Rice has high nutrient requirement, particularly Nitrogen. Wheat requires relatively lesser amount of nutrition. This allows for efficient nutrient cycling.
 - » **Irrigation facilities** at nominal charges
 - » **Assured Procurement** at a MSP
 - **Emerging Problems from RWCS**
 - » **Deteriorating Soil Health:**
 - **Excess use of Fertilizers**
 - The puddling in rice cultivation destroys soil structure, leading to soil aeration and soil compaction. This has resulted in hardpan at shallow depths that halts the root penetration/proliferation in the soil.
 - **Depletion of Soil Nutrients** in the absence of nitrogen fixing crops.
 - » **Ground Water Depletion:** Both rice and wheat are dependent on irrigation in NW India and India extracts more than 25% of world's groundwater.
 - » **Herbicide Resistance:** Continuous use of a single herbicide in rice for the control of *Echi*
 - » **Escalating Production Cost:** Cost of fertilizers, hybrid seeds etc. have kept on going up. Similarly, the cost of hiring laborers for transplantation is also going up.
 - » **Decreasing Productivity:** There is long term experiments that the productivity of RWCS is stagnating, and sometimes declining. This stagnation may be due to exhausted natural resource base along with adverse impact of climate change.
 - » **Environment Pollution and Greenhouse gas emissions** due to crop residue burning.

- » **Increased fiscal burden:** MSP for Wheat and Rice and Fertilizer subsidy (especially Urea)

- **Solutions:**

- » **Crop Diversification and Crop Rotation** can play a crucial role in efficient resource utilization and reduced wastage and pollution.
- » **Crop Residue Management** - Promote use of technologies like **in-situ residue incorporation**, zero tillage sowing of wheat, fast decomposers (e.g PUSA decomposer).
- » **Sustainable Intensification Technologies:**
 - **Conservation Agriculture** has emerged as a promising technology to get maximum efficiency of the inputs and to sustain the productivity of the RWCS.
 - It works on **three fundamental principles**, namely, Crop Diversification, Zero or reduced tillage, and residue retention. This also helps in carbon sequestration and improvement in physiochemical properties of soils.
 - **Dry direct-seeded rice (DDSR)** has evolved as a potential alternatives to puddled transplant rice.
 - Here seed of rice is sown with or without tillage and irrigation. Irrigation is applied at certain intervals.
 - **Advantages:**
 - Water Conservation
 - Reduce Methane emission
 - Early maturity of rice (7-10 days earlier)
 - **Water Saving Technologies:**
 - The tensiometer-based irrigation scheduling, laser land leveling, direct seeding of rice, subsurface drip irrigation, furrow irrigation, and alternate wetting and drying for rice or applying irrigation at hairline cracking some of the major technologies for efficient water use in RWCS.
- » **Precision Agriculture** can play a crucial role in efficient resource utilization and reduced wastage and pollution.
- » **Soil Health Improvement:**
 - The fields which witness loss of soil organic matter due to puddling can be fortified with the use of **farmyard manure**.
 - In situ management of rice stubble with microbial decomposer can also help to improve soil organic matter content.

- **Conclusion**

- » RWCS has started to face many sustainability issues. Therefore, there is a need to shift from conventional RWCS to crop diversification, conservation agriculture, and various sustainable intensification technologies and water saving technologies.

2) QUESTION-2

- **Introduction:**

- » The Rice Wheat Cropping System (RWCS), which had become the backbone of India's Agriculture system, has in recent times started facing various sustainability issues which has led to decreased yield.

- **Key factors leading to reduction in yield:** (already discussed in details in the above question)

- » **Land Degradation:**
 - Excessive use of chemical fertilizers, pesticides, insecticides
 - Puddle action at the time of rice cultivation results in formation of hardpan at shallow depth.
 - » **Nutrient Imbalance** - Excessive dependence on UREA
 - » **Development of Herbicide Resistance of weeds**→ Removing weed is becoming more and more challenging.
 - » **Climate change** is leading to increased growth of pests/insects damaging crops. It is also causing increased variability in the weather pattern, rainfall etc., impacting production.
 - » **Land Fragmentation** also leads to reduction in efficient use of resources.
 - » **Cost of input** (including labor cost) has been increasing substantially
 - » **Slow pace of technology adoption** has led to low mechanization, less use of modern agriculture technologies etc.
- **Crop Diversification** can stabilize the yields in the following ways:
 - » **Reducing the risk of soil degradation** by providing break from the continuous cultivation of rice and wheat.
 - » **Suitable Crops** for the soil type can be chosen as part of diversification. This can lead to improved yield from the soil.
 - » **Diversifying to leguminous crops** or following crop rotation approach can increase soil fertility by natural nitrogen fixation.
 - » Diversification may also break the lifecycle of pests, weeds etc., benefitting agriculture yield in long run.
 - » **Risk reduction** is another major advantage. When farmers start depending on diverse kinds of crops, damage or price fluctuation of one crop will not lead to drastic impacts on farmers.
 - » **Improved resilience to climate change** by diversifying to crops which are more tolerant to different climatic conditions.
 - In conclusion, we can say that crop diversification is a sustainable way to improve the yield and resilience of the rice-wheat cropping system. It reduces the risk of soil degradation, breaks the life cycle of weeds, pests etc and improves the resilience of agriculture system to climate change.

9. RICE: CROPPING PATTERN

- India is the 2nd largest producer of rice after China.
- **Conditions for Rice Cultivation:**
 - » **Soil type: Deep clayey and loamy soil.**
 - » **Climate: Tropical (Hot and Humid):** The temperature should be fairly high at mean monthly of 24°C. The average annual rainfall required for rice is 150 cm.
 - » It is the dominant crop in areas of over 200 cm annual rainfall and is still an important crop in areas of 100-200 cm rainfall. The 100 cm isohyet forms the limit of rice in rainfed areas. In areas receiving less than 100 cm annual rainfall, rice can be grown with the help of irrigation, as is done in Punjab, Haryana and western U.P. About 40 per cent of rice crop in India is raised under irrigation



» **Top Rice Producing states:** WB > Punjab > Uttar Pradesh > Andhra Pradesh > Bihar

- **Major Rice Producing Areas:**

- » Rice successfully grows from sea level to about 2,000 m altitude and from humid areas in eastern India to dry but irrigated areas of Punjab, Haryana and northern Rajasthan.
- » In southern states and West Bengal, the climatic conditions allow the cultivation of two or three crops of rice in agricultural year. In West Bengal, farmers grow three crops of rice called 'aus', 'aman', and 'boro'.
- » But in Himalayas and in north-western part of the country, it is grown as a Kharif crop during southwest Monsoon season.

1) DIRECT SEEDED RICE (DSR)

- **Background: Traditional method of rice cultivation and its impact:**

- » In India (including in north-west India), transplanted rice is the most pre-dominant method of cultivation. But it suffers from a lot of limitation.
 - **Requires access resources:**
 - It requires around 25 cm of water for puddling operation, which creates a dense clay layer in the sub-soil to prevent seepage losses. Further, the crop requires around 130 +/- 10 ha-cm of irrigation; suitable variety of seeds; and recommended dose of fertilizers.
 - The amount of water required is so high that around 40% of all irrigation water used in north-west India goes for paddy cultivation.
 - **Causes global warming:** Further, flooded rice field produce around 10% of global methane gas.
 - **Causes nitrogen pollution:** Injudicious fertilizer use also lead to nitrogen pollution and nitrous oxide emissions.
 - **High Labour dependency** especially during the transplantation phase.

- Considering the limitations of transplanted paddy, scientists have developed several alternative **direct seeding methods**.

- » In this method, there is no requirement of raising nursery for transplantation.
- » Here paddy seeds are drilled into the field with the help of a machine (zero till drill) that does seeding of rice and spray the herbicide simultaneously.
- » **Weed Management** plays a significant role in DSR as the technique doesn't require flooded fields and thus weeds tend to grow easily, unlike the conventional method. So, weedicide/herbicide is applied both at pre-emergence and post-emergence stage (25-35 days after sowing).
- » **Advantages:**
 - **Less labour requirement**
 - **Lesser water requirement** (upto 25% lower) compared to transplanted flooded rice.
 - Less water also reduces energy demand for pumping of irrigation water.
 - **Lesser seed requirement:** Around 20-25 kg/ha compared to 60-80 kg/ha required for transplanted paddy.
 - **Enhanced fertilizer use efficiency** due to placement of fertilizer in the root zone.
 - **Early maturity** (by 7-10 days). This allows timely sowing of succeeding crops.
 - **Reduction in carbon footprint** (by reduction of methane emission)
 - **Little disturbance to soil structure**

- **Higher productivity** of both paddy and wheat by 5-10%.
- **Some Limitations of DSR:**
 - » Laser and Levelling are compulsory - the field should be laser levelled
 - » High herbicide dependency
- **DSR method is relatively more popular in the rainfed rice growing states like Chhattisgarh.** There is a scope to upscale the technology in the North-West and Indo-Gangetic plains.
- Government of Punjab has also taken steps to promote DSR but hasn't been successful. This is because farmers needed a well moist soil before DSR but several factors like, High Temperature, Deficient Rainfall, erratic power cuts, inadequate canal water availability (owing to the breach in Sirhind Canal in May) etc. prevented this.
- **Conclusion:**
 - » Though alternatives like DSR can reduce water use efficiency of paddy, but the requirement of water even in this method is pretty high. States like Punjab, which depend on irrigation for rice cultivation should try incentivizing farmers to move away from rice cultivation.

10. TEA

- **Introduction**
 - » Tea is the dried leaf of a bush. It contains theine (caffeine) and when added to boiling water along with sugar and milk, gives a stimulating drink. It is the most important beverage crop of India.
- **Tea Cultivation**
 - » **Requirements for Tea Cultivation**
 - **Climate:**
 - Tropical and subtropical climate endowed with deep and fertile well drained soil, rich in humus an organic matter.
 - A moderately hot and humid climate, which is preferred for better yield, crop distribution and quality
 - Frequent showers evenly distributed over the years ensure continuous growth of tender leaves.
 - **Temperature:** An ambient temperature within 13 degree Celsius and 28-32 degree C is conducive for growth of tea. Maximum ambient temperature above 32 degree C is unfavorable for optimum photosynthesis. Heat is synergically disastrous for the crop if it is accompanied by low humidity.
 - **Winter Dormancy:** At temperature below 12 degree C, there is hardly any growth during this period. Flushing in the tea plant starts from March, with the rise in temperature.
 - **Soil**
 - Acidic soil with around 4.5-5.5 pH. Suitable soil for tea cultivation is well drained, deep, friable loamy soil. **The best soil for tea cultivation is forest soils rich in humus and iron content**.
 - **Other Requirements**
 - Cheap and efficient labour as tea is almost exclusively hand picked.
 - Tea grows better when planted along the shady tree.
- **Major tea producing area in India**

- India's major Tea producing areas

- The main tea growing regions are in Northeast India (including Assam) and in North Bengal (Darjeeling district and the Dooar region). Tea is also grown on large scale in Nilgiris in South India.
- **Assam:** Darrang, Goalpara, Kamrup, Lakhimpur, Dibrugarh, Nowgong, Sibsagar, Cachar, Karbi Anlong, North Cachar.
- **West Bengal:** Darjeeling, Terai (west Dinajpur), Doors (cooch Bihar)
- **Tamil Nadu:** Kanyakumari, Tirunelveli, Madurai, Coimbatore, Nilgiris.
- **Kerala :** Cannanore, Palghat, Kozhikode, Malappuram, Trichur, Trivandrum, Quilon, Kottayam, Ernakulam, Idukki, Wayanad.
- **Karnataka:** Chikkamagaluru, Coorg, Hassan.
- **Himachal, Uttarakhand, Meghalaya, Andhra Pradesh, and Tripura** are also tea producing states in the country.



- In 2022, India produced 1336 million kgs of Tea.

- » **Assam** produces around 55% of the tea and WB another 30%.
- » Tamil Nadu, Kerala and Karnataka were other major producers

- Tea Consumption in the country

- » India is also one of the largest tea consumers in the world, without about 3/4th of the total production consumed in the country itself.

- Exports

- » India exports CTC (crush-tear-curl) grade mainly to Egypt, Pakistan and the UK, and the orthodox variety to Iraq, Iran and Russia.

- Key problems faced by the Organized Tea Industry in the country (not exactly in Syllabus)

- » **Reducing Production:** Climate change, sweltering heat etc in Brahmaputra valley Assam, and West Bengal have adversely impacted production.
- » **Scarcity of labor and its cost** (65% of the cost) in the organized industry
- » **Stagnant Prices:** Prices haven't increased in tandem with inflation.
- » **Increased competition:**
 - **Emergence of the small tea growers** as a dominant force in the industry
 - Lately, the organized sector's production has shown a declining trend and small tea growers now have a larger share of the pie.
 - **Influx on tea from Nepal** also adding to Darjeeling woes.

» **Deteriorating quality due to aging tea bushes** in this centuries-old industry.

- **Way Forward**

» **Steps government can take:**

- Regulating tea waste being sold at lower prices in domestic market.
- Restricting import of poor-quality tea
- Promoting consumption by emphasizing on its health benefits.

» The industry has to gear up to face its challenges.

- Adopt mechanization
- Diversify export destinations

11. COFFEE

- **Example Questions**

- » Discuss the cropping pattern of Coffee in India. What are the key climatic condition requirements for coffee production [10 marks, 150 words]
- » Discuss the key challenges faced by Coffee sector in India [10 marks, 150 words]

- **Introduction**

- » Coffee is a tropical plantation crop. Its seeds are roasted, ground and are used for preparing beverage. In India, it is the second most important beverage crop, next only to tea.
- » It is indigenous to Ethiopia. In India, Coffee plantations started in 18th century. Over the years, we have earned a distinct identity for our coffee. India is the only country in the world where coffees are grown under a 'well-defined two-tier shade canopy' of evergreen leguminous trees.
- » There are three varieties of coffee i.e. **arabica, Robusta and liberica**. India mostly grows the superior quality coffee, arabica, which is also in great demand in the international market.
- » India is one of the top 10 coffee producers in the world (generally ranked 5th or 6th) -> After Brazil, Vietnam, Columbia and Indonesia. We produce around 5% of the world coffee.

- **Total Production and Exports**

- » India's coffee production is estimated to rise by 15% in 2022-23 from a year before to 393,400 tonnes.
- » India exports 80% of its production to more than 40 countries.
 - The exports reversed a COVID-induced slide to record a 42% year on year jump in 2021-22, exceed the \$1 billion mark for the first time.

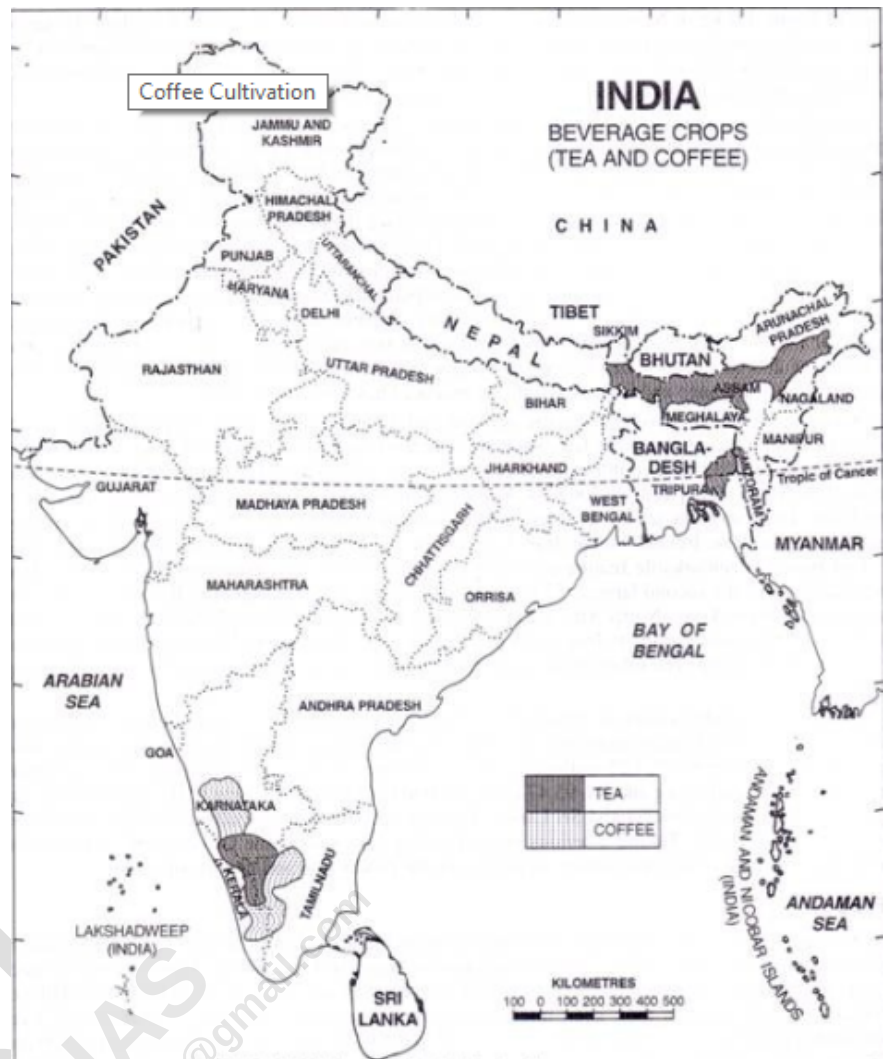
- **State wise production**

- » Coffee plant was grown for the first time on Baba Budan Hills (Karnataka) in India.
- » The **three southern states** account for almost all of the coffee production in the country with **Karnataka (71%)** being the largest producer followed by Kerala (21%) and Tamil Nadu (5%).
- » In Karnataka, the Kodagu and Chikkamagaluru district account for over 80% of the state's total output.

- **Climatic Condition Requirements**

- » Since coffee is a **tropical plantation**, it requires **hot and humid** climate with temperature varying between 15 degree Celsius to 28 degree Celsius and a rainfall between 150-250 cm.

- » It grows on the hilly slopes at the height of 900-1800 m. These conditions are prevalent in the hill areas of Nilgiris where the coffee plantations are mostly confined.
- » **Dry weather** is necessary at the time of ripening of the berries.
- » It doesn't tolerate frost, snowfall, high temperature above 30 degrees Celsius and is generally grown under shady trees.
- » **Cheap and skilled labor force** is another requirement for coffee cultivation which is required in sowing, transplanting, pruning, drying, grading and packing of coffee.



- **Coffee Industry is going through a Crisis:**

- » **Climate Change:** Erratic rainfall and rising temperature is disrupting ideal growing conditions.
 - for e.g. Dry spells in 2015 and 2017 and floods and landslides in 2018 and 2022
- » **Pest and Diseases:** Erratic weather conditions are helping pests to breed and new diseases to emerge.
 - Farmers have reported fruit rot, stalk rot, root rot and other irreparable damages due to heavy rainfall.
- » **High Cost of Finance** for coffee growers when compared to their international counterparts. This is hampering India's export competitiveness.
- » **Increasing cost of input** (fertilizers, agrochemicals, diesel (for irrigation) labour cost etc.)
 - For e.g., in Brazil labour cost account for 25% of the cost of production, whereas in India it accounts for 60% of the cost of production.
 - Mechanization is difficult in India as the Indian Coffee is grown on elevation and shade grown.
- » **Difficult to find skilled workforce:** Traditional workforce is moving to cities in search of better opportunities and migrants are not well trained for the coffee plantation work.
- » **High NPAs and threats of proceedings under SARFAESI Act**

- » **Poor brand recognition** of Indian coffee in global market: Indian coffee is of very high quality but the global customer is not aware of this coffee's Indian origin. This is despite Indian coffees offering innumerable brews, flavours, aromas and blends.

- **Way Forward:**

- » **More R&D Support** -> Development of flood resistant and drought resistant variety.
- » **Interest Subvention initiatives** for coffee growers to reduce the cost of finance.
- » **Brand Creation for Indian Coffee** in global market
 - India's coffee variety is unique. It is the only country to produce shade grown coffee.
 - Coffee Board of India is working on a separate India brand and certification system for coffee growers.
- » **Increasing domestic demand** for Indian Coffee
- » **Finding alternative source of revenue for India's coffee growers.**
 - Intercropping with non traditional crops such as fruit bearing trees, food crops etc. can be beneficial
 - Note: traditionally coffee was intercropped with pepper and cardamom.
 - Apiary or green tourism can increase the income of coffee growers.

- **Conclusion:**

- » Government is moving in right direction through the Coffee (Promotion and Development Bill), 2022. Coffee Board needs to ensure that Coffee growers and Coffee industry gets all the support which other farmers are able to secure from government.

12. PULSES

1) QUESTIONS

- Discuss the key causes and consequences of stagnation in acreage, production and productivity of pulses in India over the last three decades. [15 marks, 150 words]
- Discuss the key strategies that can increase domestic production of pulses. [10 marks, 150 words]
- **Intro:** Pulses are the important source of proteins, vitamins and minerals and are popularly known as "Poor man's meat".
- **Advantages Positives about Pulses**
 - **Nutritional Security:** In a country like India, where many people are poor and vegetarian, pulses are an important and affordable source of protein.
 - **Suitable for Marginal Environment:** Drought resistant and deep rooting species of pulses can supply ground water to companion crops when planted in inter cropping pattern.
 - Locally adapted pulse varieties can enhance production system in dry environments.
 - **Increase fertility of Land:** The leguminous plants of pulse also help in nitrogen fixation and thus ensuring higher fertility of soil.
 - **Low food wastage footprint:** Pulses can be stored for longer period without losing their nutritional value and minimizing loss.

- **Various pulses and production in India.**
 - India is the **largest producer** (25% of global production), **consumer** (27% of world consumption) and **importer (14%)** of pulses in the world.
 - They account for 20% of India's total area under cultivation and provide 7-10% of the total food grains in the country.
 - **The overall pulse production in India** has gone up from 8.4 million tonnes in 1950-51 to 26 million tonnes in 2022-23.
 - In fact, in the last decade, India's production has increased by 50% (from 18 million tonnes to 27 million tonnes)
 - But, pulse production has not increased in step with the population growth, per capita availability of pulses have declined from 22.1 kg per person in 1951 to 16.4 kg per person in 2022.
 - Though there is surplus production of Chana, the imperfect substitution among pulses and limited international availability put pressure on the prices of some pulses.
- **Major Pulse Producing States:**
 - Madhya Pradesh, Maharashtra, Rajasthan, UP, and Karnataka.
- **But, the per capital availability of pulses have been decreasing in India.** The primary reasons for domestic shortage of pulses are:
 - A. **The increase in area under cultivation, production and productivity of pulses has been extremely slow**
 - The production of pulses grew only by 45% from 1951 to 2008, while wheat production increased by 320% and rice by 230 percent.
 - India has been import dependent since 1981.
 - B. **Low Yield, increased irrigation facilities and Blue Bulls trouble**
 - C. **Open ended procurement of wheat and rice under MSP -> Lack of assured price for pulse**
 - D. **Very less R&D on pulses globally** (due to very less consumption in advanced western countries)
- **Steps which have been taken to increase pulse production**
 - A. **Measures to incentivize Pulse Production** under National Food Security Mission (NFSM), Minimum Support Price Programs and by Increase production.
 - For e.g. PM AASHA's prize support scheme specifically focused on increasing the procurement of pulses on MSP.
 - B. MoA&FW have also formulated a Special Kharif Strategy which was implemented during Kharif 2021:
 - Here, HYVs of seeds available with Central Seed Agencies or in the States were distributed free of cost to increase area under pulse production through intercropping and sole crop.
 - C. **Crop Diversification Program** (a sub scheme of RKVY) is being implemented in original green revolution states viz. Punjab, Haryana, and in Western Uttar Pradesh to diversify paddy areas towards less water requiring crops like oil seeds, pulses, coarse cereals, agro forestry etc.

- **Way Forward:**
 - » **Augmenting domestic production**
 - **Policy reforms** like higher MSP for pulses and inclusion of pulses under the PDS system.
 - **Improved Agri-Inputs for Pulses:**
 - **Increased R&D** to focus on developing new HVYs which are adapted to drought, climate change etc.
 - **Promoting** cultivation of pulses in well irrigated areas and ensuring better irrigation facilities in existing pulse growing areas.
 - **Ensure timely availability of chemical fertilizers** which has remained a problem for this sector.
 - Similarly, inadequate availability of Gypsum or pyrites as a cheap source of sulphur remains a serious impediment in many states.
 - **Prioritizing the crop insurance scheme in pulse growing regions and its proper implementation**
 - **Promoting Scientific Cultivation:** Currently, improper sowing time, low seed rate, defective sowing methods, insufficient irrigation etc. are the major agronomic constraints.
 - » **Focus on solution for blue bulls trouble**
 - Biodiversity protection and conservation in forest areas could ensure enough availability of food for blue bulls in the forest areas and reduce their infiltration in cultivated areas.
 - » **Improving technology of Dal Mills -> Reduce wastage**
 - Most of the dal mills have out dated technologies, resulting in excessive loss of pulses in the form of broken or powdered grains.
 - » **Distribution channels** also have to be made more efficient.
 - » **Dealing with fluctuation of prices** - Increased buffer stocks and diversifying imports.

13. MILLETS

- **Question:**
 - » Discuss the potential of millets in climate resilient agriculture and the conservation of agrobiodiversity [10 marks, 150 words]
 - » How has the emphasis on certain crops brought about changes in cropping patterns in recent past? Elaborate the emphasis on millets production and consumption. [Mains 2018, 15 marks, 250 words]
- **Introduction (Current Affairs/Context Based)**
 - » **International Year of Millet (IYM):** The United Nations General Assembly has declared the year 2023 'International Year of Millets'. It will help in creating awareness throughout the world about the significant role of millets in sustainable agriculture and its benefits as a smart food and superfood.
 - » IYM 2023 aims to contribute to the UN 2030 Agenda for Sustainable Development, particularly SDG 2 (Zero Hunger), SDG 3 (Good health and well being), SDG 8 (Decent work and economic growth), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate action) and SDG 15 (Life on Land)



- **Definitions:**
 - » Millets include Jowar, Bajra, Ragi, little millets including Kutki, Kodo, Sawa, Kangni and Cheena.
- **Cropping:**
 - » They are generally cultivated in low-fertile land, mountains, tribal and rain-fed areas.
 - These areas include Andhra, Chhattisgarh, Gujarat, Haryana, MP, Rajasthan, MHA, KAR, UP, TN and Telangana.
 - » India is the **largest producer and second largest exporter of millet** and in 2022 India produced around 50.9 million tonnes. This accounts for 80% of Asia's and 20% of global production. India is followed by African countries like Nigeria and Niger in production.
 - » **Decreased Production over the years:**
 - In pre green revolution era (1965-66), millets were cultivated in 36.90 million hectares of the country. In 2016-17, the area reduced to 14.72 million hectares.
 - **Why?**
 - **Green Revolution** increased the productivity of wheat and rice.
 - Expansion in irrigation.
 - MSP Policy
 - **Changes in consumption pattern, dietary habits etc:** Socio-economic dynamics resulting from the hardy nature of the crop, relegated them to be the grain of the poor.
 - **But recent studies have highlighted various significance of millets for healthy life and sustainable economic development:**
 - » **Agri-Sustainability:**
 - **Climate Resilience:** Millets are tolerant to droughts, intensive to excess sunlight etc.

- **Water Efficient:** Millets can survive in less water conditions and can solve the problem of over-extraction of water resources.
 - A new study published in the journal ***Nature Water*** in Oct 2023 highlights that shifting to millets increases groundwater recharge more than drip irrigation in India's northern plains.
- **Better Health:**
 - » **Food Security:** In arid areas, millets are often the only crops that can be harvested in the dry regions and are a crucial part of household food basket.
 - » **Nutrition:** Millets are smart food which are rich in nutrients like protein, vitamin-A, iron, calcium, iodine etc.
 - For e.g., just 100 gm of daily cereals (rice) intake with finger millets (ragi) will increase the daily iron intake by 50% and calcium by 350%.
- **Government Initiatives to promote Nutri-Cereals:**
 - » **Union Budget for FY24** announced an initiative focused on '**Making India a Global Hub for Millets**' (Shree Anna).
 - The Indian Institute of Millet Research, Hyderabad, will be supported as the Centre of Excellence for sharing the best practices, research and technologies at the international level.
 - » **MAHARISHI Initiative** i.e. Millets and Other Ancient Grains International ReSearch initiative. This international initiative will focus on research and awareness via agro-biodiversity, food security and nutrition aligning with the International Year of Millets.
 - » **India had declared year 2018** as the **Year of Millets:**
 - Spreading **awareness** about nutritional benefits of nutrients which will help in increasing the demand resulting in remunerative prices for poor and marginal farmers.
 - Under the **Sub Mission on National Food Security Mission (NFSM) - Nutri Cereals** is creating awareness among farmers for Nutri Cereals (Millets).
 - NFSM - Coarse Cereals are divided into two components
 - NFSM (Makka and Jau)
 - Sub Mission on Nutri-Cereals covering Jowar, Bajra, Ragi and little millets like Kutki, Kodo, Sawa, Kangni and Cheena
- Although significant strides have been made by many stakeholders in the promotion of millets there are **still several aspects that need to be strengthened** from both the demand as well as supply side.
 - **Increasing Demand:**
 - i. **Consumer Awareness:** To increase demand and make them a regular food option, mission mode campaigning is required which not only encourages people to move towards millets but also counters myths and misconceptions as well as demystifies their cooking.
 - ii. **Promote startups** supporting innovative and functional products with millets.
 - iii. **Gluten Free** value added products made from millet can be developed for the export market.
 - **Supply Side:** Production as well as Processing needs to be supported and encouraged in different states.

- i. **Revise traditional methods** of cultivation.
 - For e.g. Systems like Barahnaja (twelve seeds) from Uttarakhand and other mixed cropping practices in different states not only contributed to food security and soil fertility but also to diet and nutrition diversity by including millets, legumes and other nutritious crops in diet.
- ii. **Provide increased support to farmers** - Increased procurement under MSP
- iii. **Increased market linkage** can encourage farmers to grow millets.
- iv. **Increased R&D** for development of High Yielding Varieties.

14. OILSEEDS AND VEGETABLE OIL

1) QUESTION-1

- **Ques:** Examine the major challenges faced by oilseeds farmers in India. Analyze the scope of increasing oilseed production through research and development, technological advancements, and agronomic practices. [15 marks, 250 words]
- **Oilseed Situation in India:**
 - » Area under Oilseeds cultivation has increased from 10.7 million hectares in 1950-51 to **28.8 million hectares in 2020-21**. The **production of Oilseeds** has also increased to 36.1 million tonnes in 2020-21. But, even now, **India imports majority of it's edible oil consumption**.
 - » **The domestic availability of edible oil** has risen from 86.30 lakh tonnes in 2015-16 to 121.33 lakh tonnes in 2023-24 (ESI 2023-24)
 - **Percentage share of imported oil** has reduced from 63.2% in 2015-16 to 57.3% in 2022-23.
- **Reasons for Low Oil Seeds Production in India:**
 - » **More Focus on Wheat and Rice:** Green Revolution, MSP Regime, High Yielding Varieties etc.
 - » **Productivity and yield of oilseeds crop** is low.
 - **Palm oil** which has high productivity hasn't grown in India
 - Lack of access to **improved variety of seeds** (for e.g. GM varieties like DMH-11 are still not allowed to be grown commercially.
 - Specialized inputs like specific fertilizers, pesticides etc. are not easily available.
 - » **Poor Infrastructure:** Inadequate irrigation facilities
- **Government Initiatives** to reduce the import dependency has been focused on increasing the production and productivity of oilseeds:
 1. **National Mission on Edible Oils – Oil Palm (NMEO-OP)** aimed at increasing availability of edible oil in the country by harnessing area expansion and through price incentives.
 2. **Improving the Seed Varieties: National Food Security Mission – Oil Seeds (NFSM-Oilseeds)**
 - Focused on **production of foundation and certified seeds** and **distribution of certified seeds and seeds mini kits of latest high yielding varieties**.
 - Under NFSM-Oilseeds, government of India has set up **36 oilseeds seed hubs** during 2018-19 and 2019-20 with **an objective to increase the availability of high yielding quality seed**.

3. **Incentivization of oilseeds** through **MSP regime**: For e.g. A remunerative MSP for rapeseed and mustard (which was 98% over the cost in 2022-23) is also providing the incentive to farmers to diversify production.
 4. **Atmanirbhar Oil Seed Abhiyan** (announced in interim Budget 2024-25)
 - Under this a strategy will be formulated to achieve 'Atmanirbharta' for oil seeds such as mustard, groundnut, sesame, soybean, and sunflower. The initiative will focus on research for high-yielding varieties, widespread adoption of modern farming techniques, market linkages, procurement, value addition, and crop insurance.
 5. Other steps like Pradhan Mantri Krishi Sinchayi Yojna, MSP for Oil seeds, subsidies for inputs such as fertilizers and pesticides, Training and extension programs for farmers etc. also help oil seed sector.
 6. **Asian Palm Oil Alliance (APOA): 5 Major Palm Oil Importers from Asia form alliance (Sep 2022)**
 - The apex edible oil industry associations from five major palm oil importing countries of Asia - India, Pakistan, Sri Lanka, BD, and Nepal - have come together to form the Asian Palm Oil Alliance (APOA), at the instance of Solidaridad Network, a specialist in sustainable agriculture. The purpose is to safeguard the economic and business interests of the palm-oil consuming countries.
- **More Steps that could be taken:**
- » **Research and Development:** To develop newer, high yielding varieties of oil seeds.
 - E.g. researchers at International Crop Research Institute for the Semi-Arid Tropics (ICRISAT) have developed a new variety of groundnut that is resistant to ground nut rosette virus.
 - » **Technological advancements:**
 - Advancements like precision agriculture, data analytics etc. can improve our understanding of crop requirements and regions where oilseeds will be more suitable.
 - Improved techniques of irrigation like drip irrigation, sprinkler irrigation can allow efficient use of water and makes it possible to grow oilseeds in areas with limited water resources.
 - » **Better Agronomic Practices:**
 - Techniques like inter-cropping, crop rotation etc. can help in increasing area under cultivation for the production of oilseeds, and improve the overall oilseed yields.
 - like precision agriculture, data analytics, micro-irrigation t can help us better understand region suitable for oilseeds and the kind of fertilizers, insecticides etc. required.
 - » These improvements together can also help in making agriculture climate resilient.
- These methods, along with policy support like NMEO-OP and extension services can help farmers move towards increased oilseed production and can help India become Atmanirbhar in oilseed production.

2) QUES-2

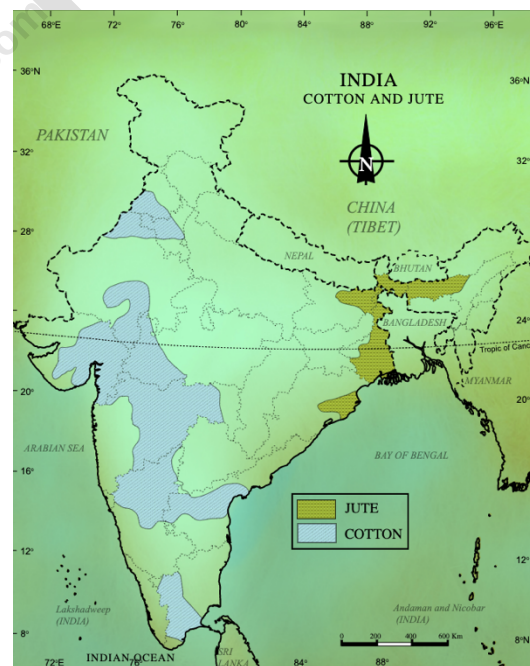
- **Question:** Discuss the potential of oil-palm cultivation in India and its role in meeting the domestic demand for edible oils. How can National Mission on Edible Oils - Oil Palm (NMEO-OP) help in achieving this potential? [15 marks, 250 words]
- **Introduction:** (Talk about Edible oil shortage in India and import dependency)
- **Oil Palm Cultivation in India has significant potential due to following reasons:**

1. **Increasing Demand of vegetable Oil:** With increasing population and increasing urbanization, the demand for vegetable oil is going to increase in the country.
 2. **Agro-Climatic Conditions:** Oil Palm can be grown in tropical and subtropical climates of India. Regions like Tamil Nadu, Kerala, Karnataka, and North-eastern India are well suited for palm oil cultivation.
 3. **Potential to reduce India's import dependency:** Of all the vegetable oils imported in India, Palm oil constitute around 60%.
 4. **High Yield of palm oil:** It produces 10 to 46 times more oil per hectare compared to other oilseed crops.
 5. **Other value added products** - Other than cooking oil, Palm Kernel oil is used in cosmetics and pharmaceutical products.
 6. **Increased economic opportunities and Jobs:** Large scale oil palm cultivation can lead to the establishment of oil processing industries, creating a value chain that would enhance economic activities. It is also a labor intensive sector and would enhance economic activities and local development.
- **However**, introduction of largescale oil palm cultivation has also been associated with some challenges:
1. **Deforestation and biodiversity loss:** Indonesia has seen large scale deforestation and biodiversity loss to expand oil-palm cultivation
 2. **High Initial Cost:** The oil palm tree takes 3-4 years to mature and start producing oil palm. This can be barrier for most Indian farmers.
 3. **Price Fluctuation** is also a major concern which limits the scope of
 4. **High Irrigation requirement:** Various parts of the country are already suffering from water stress and ground water depletion.
 5. **Shortage of planting material.**
- **National Mission on Edible Oil - Oil Palm (NMEO-OP) aims to reduce some of the above challenges:**
- Launched as part of Atmanirbhar Bharat Scheme, **it aims to expand India's oil palm cultivation from around 3.5 lakh hectares to 10 lakh hectares** in a phased manner. It also targets to increase crude oil production **to 11.2 lakh tones by 2025-26 and upto 28 lakh tonnes by 2029-30.**
 - **Budgetary allocation** - Rs 11,040 crores (of which Rs 2,196 crore will come from states)
 - **To tackle High Initial cost**, a substantial increase in assistance for inputs/interventions has been made:
 - **Substantial increase in assistance for inputs/interventions:**
 - For Planting Materials for oil palms, a substantial increase from Rs 12,000/ha to Rs 29,000/ha has been made.
 - Substantial increase has been made for maintenance and intercropping interventions -> A special assistance @Rs 250 per plant is being given to replant old gardens for rejuvenation of old gardens
 - **To tackle price fluctuation**, government is also providing **price assurance to palm oil farmers** for the fresh fruit bunches. This is called **viability price**. It will protect farmers from price fluctuations in international market.
 - The VP will be annual average CPO price of the last five years adjusted to WPI Index, multiplied by 14.3%.
 - There is a sunset clause for the scheme which is 1st Nov 2037.

- The Scheme will have special focus on north-east region and the **Andaman and Nicobar Islands**.
 - To give impetus to the NE region and Andaman, the government will additionally bear a cost of 2% of the CPO price to ensure that the farmers are paid at par with the rest of India
 - To address the issue of shortage of Planting Material in the country, **seed gardens will be provided assistance upto Rs 80 lakhs for 15 ha** in rest of India and Rs 100 lakhs per ha in NE India and Andaman region.
 - **For Environment sustainability** NMEOP advocates responsible cultivation practices that align with India's National Biodiversity Action Plan. It also has provisions for investment in R&D for the development of drought resistance varieties.
- Thus, NMEOP, with its comprehensive and multi-faceted approach, holds promise in realizing the potential of oil-palm cultivation in India, helping meet domestic demand for edible oils sustainably and equitably.

15. COTTON

- **Example Questions**
 - » What are the key climatic requirements of Cotton cultivation. How is cultivation of cotton distributed geographically in the country? [10 marks, 150 words]
- Cotton is one of the principal commercial crops of India and India accounts for around 25% of the total global cotton production. It plays a major role in sustaining the livelihood of an estimated 6 million cotton farmers and 40-50 million people engaged in related activity such as cotton processing & trade.
 - » Due to its economic significance, in India, it is also termed as "**White Gold**".
- **Suitable Climate Condition for Cotton**
 - » Cotton is a crop of tropical and subtropical areas and requires uniformly high temperatures between **21 degrees and 30 degrees C**. The growth is negatively impacted if the temperature falls below 20 degree C. Frost is harmful for the crop.
 - » The crop has modest water requirement (average annual rainfall of **50-100 cm**) and can be grown in areas with lower rainfall with the help of irrigation.
 - » **Good sunshine** is a must at the time of flowering and moist weather or heavy rainfall at the time of ball opening and picking are detrimental to the crop.
 - » **Other requirements:** Cheap and skilled labor force at the time of picking of cotton. Normally the picking season is spread over a period of 3 month.
- **Traditionally**, it is cultivated on the **lava plateau of Deccan** and therefore the soil here is called the **Black Cotton soil**.
- There are **three major cotton producing regions in India**:
 - i. **Northern Zone:** South East Punjab, Western Haryana, and Northern Rajasthan
 - ii. **Central Zone:** Gujarat, Maharashtra, Western Madhya Pradesh and neighboring Southern Rajasthan in Western India
 - iii. **Southern Zone:** Telangana, Andhra Pradesh, North Karnataka and Tamil Nadu
- **Initiatives: Budget 2023-24:** To enhance productivity of extra-long staple cotton, we will adopt **a cluster based and value chain approach through Public Private Partnerships (PPP)**. This will mean collaboration between farmers, state and industry for input supplies, extension services, and market linkages
- **Conclusion**
 - Cotton cultivation, together with cotton textile industry employs millions of people in India and therefore can play a very important role in improving the employment situation in the country. In order to increase the attractiveness of India's cotton and cotton products, Ministry of Textile has recently released brand and logo for Indian Cotton - **Kasturi** - on the occasion of 2nd World Cotton Day on 7th of October 2020.



1) PROBLEM OF DEVELOPMENT OF RESISTANCE IN PESTS SUCH AS PINK BOLL WORM (PBW), TO BT COTTON HAS POSED A CHALLENGE TO SUSTENANCE OF BT-COTTON

- **Background:** Indian farmers have faced consistent loss of Bt Cotton crops due to pink bollworm attacks since the mid-2000s, when scientists found that the insect had become resistant to the genetically modified variety of cotton.
 - **About PBW:** PBW is a worm that destroys parts of the developing cotton fruit, such as the square (flower bud) and the boll (rounded sac of seeds with cotton fibres).
- **How was resistance developed by PBW: Key Factors:**
 - **Early Sowing and Late Sowing:**
 - The early sowing season coincide with the time the PBW comes out of hibernation or the diapause stage in the winter months. The pest survives in this stage between two cotton seeds or cotton crop residue.
 - The cotton plants are at bud or flowering stage, during which the PBW searches for food and begins feeding on bolls during the larval stage, which continues for 14-17 days. It eventually starts laying eggs.
 - The issue worsens for farmers who sow late. The process enables worms to access food for longer periods and increasing generations.
 - The **longer duration of cotton varieties** in the south and central India, which lasted upto 150-160 days, helped the pest develop resistance to the genetically modified varieties.
 - **Not Planting other varieties against Advice:** Farmers were repeatedly advised to plant indigenous, hybrid varieties of cotton alongside Bt to prevent developing resistant. "The crossbreeding of pests from different varieties of plants would have prevented developing tolerance for longer years". But farmers haven't followed the advice.
- **Cotton Crops across the North-Indian States**, Punjab, Haryana, and Rajasthan, are reporting a severe pink bollworm attack and even Bt-Cotton is falling prey to the pest it was created to resist.
- **Impact:**
 - **Damage to crops:** Damage in 2023, is the highest since 2001 - both according to government and farmers.
 - **Note:** Before 2001, the American bollworm created havoc and ruined lives of farmers.
 - **Difficult to find laborers:** As laborers refuse to pick leftover crop as yield is too low
 - **Difficult to find buyers** as traders refuse to buy citing poor quality.
 - **Farmer Suicide:** In Sep 2023, Sri Ganganagar district saw first farmer suicide in over a decade. It was due to the fact that farmer had a lot of debt accumulated due to loss of cotton crop consecutively for 3 years.
 - **Farmers giving up cotton cultivation** in the northern belt (for e.g. the production of cotton in Punjab has almost halved in the past decade).
- **Way Forward:**
 - » Effective monitoring and timely adoption of a crop-window based integrated pest management (IPM) is the immediate solution.
 - » Scientists are also experimenting with a technology that has proven successful in the USA. This is called **PBKnot** or PB Rope L.
 - It involves using a 20 cm polyethylene hollow rope that is impregnated with pheromone gossyplure - chemical emitted by female moths - to attract male moth. This confuse the

male moth and prevent them from reaching the actual female moth. This thus disrupts the mating process and reproductive cycle.

- » **Continue the fight against climate change:** A study published in the journal Nature shows that the total developmental period of Pink Bollworm shortens with an increase in temperature and the survival rate of larvae also increases.

- **Conclusion:**

- » While promoting Crop-Window Integrated Pest Management, the GoI also needs to promote PBKnot a biotechnology tool and implement it in all cotton-producing states.

16. HORTICULTURE

1) TOMATO

• **Example Question:**

- Discuss the key factors responsible for annual increase in prices of Tomato in India. How can sustainable production and technological interventions be leveraged to enhance tomato production while addressing the concerns of both farmers and consumers? [15 marks, 250 words]

- **Introduction**

- Among the vegetables consumed in India, Tomato ranks 3 after potato and Onion, but globally it is the 2nd most consumed vegetable after Potato.
- In terms of area under tomato cultivation and in terms of total production, India ranks 2nd in the world.
 - The major tomato producing countries in the world are China, India, USA, Turkey and Egypt.

- **About Tomato production in India**

- **India's total tomato production** is around 20 million tonnes. It peaked in 2019-20 at 21.187 million tonnes and has been **declining since**. The production in 2021-22 dropped to 20.69 MT and 20.62 MT in 2022-23.
- It is typically a 90-100 day crop that starts yielding fruits 60-70 days after transplantation.
 - i) The seeds are first sown in nursery beds to raise seedlings that are transplanted in fields after around 25 days.
 - ii) Production happens in flushes.
- There are two major crops of tomato annually - Kharif and Rabi.
- There are two main crops of tomato grown in the country.
 - i) The first one transplanted from around mid-June in Central and South India (places such as Shivpuri, Sagar in **MP**, Nasik in **MHA**, Madanapalle in **AP**, Kolar and Mysore in **Karnataka** and Dindigul in **TN**) and mid-July to Aug in North India (Jhalawar and Jaipur-chomu belt in Rajasthan; Sonabhadra, Varanasi, Lucknow, Bareilly and Agra in Uttar Pradesh) and stretching to end of Sep in Eastern India (Purulia in West Bengal, Buxar in Bihar and Ranchi in Jharkhand)

- » **The autumn to late kharif crop supplies the market from Sep onwards.** This along with a smaller rabi crop transplanted during October-November, contributes to the familiar low tomato prices through the winter.
- ii) The second main crop is **transplanted during January-February.** This is a longer duration crop typically taking 130-150 days, yielding an average of 25 tonnes per acres.
 - » This is the **summer tomato** as it is **harvested during May-July** is grown mostly in regions where maximum temperature don't go beyond the mid-to-late thirties range during the flowering and fruiting season.
 - » Such conditions are mostly found in relatively cool or hilly areas such as Madanapalle, Mysore, Kolar in Karnatka; Sangamner and Narayangaon in Maharashtra, or **Solan and Mandi in Himachal Pradesh.**
- **Why increase in Prices:**
 - » **Dip in overall tomato production due to:**
 - i. **Lower acreage of tomato**
 - **Low commercial realization of the crop for farmers in the months of June in past.**
 - ii. **Extreme Weather Conditions**
 - **Heatwaves and High temperatures** in April and May along with delayed Monsoon showers in southern India and Maharashtra led to attack on tomato crops.
 - Farmers in Maharashtra have said their tomato crop was impacted by attacks of the Cucumber Mosaic Virus (CMV) and growers in Karnataka and other South Indian States have blamed the Tomato Mosaic virus (ToMV) for crop loss.
 - Later, **incessant rains** in tomato-growing regions further affected the new crop and also made transportation to non-growing regions difficult. (For e.g. rainfall in Himachal Pradesh)
 - iii. **Seasonal Fluctuation:** July and August are the lean tomato production.
 - » **Other general challenges:**
 - **Perishability of tomato** is much higher than Onion and Potato.
 - **Supply chain issues** in transporting the vegetable from areas where it is grown to regions where it is not compounds the problem.
- **Way Forward for dealing with Price Fluctuations in Tomato:**
 - » **Improved Supply Chain** - to deal with wastage due to perishability
 - Analyze why initiatives like TOP scheme and Operation Greens has failed and what lessons can be learned from the scheme.
 - Government should identify pre-existing clusters and invest in infrastructure in these areas. This infrastructure could include cold storages at the block or village levels, supported by solar dryers and cottage level processing plants.
 - Similarly, improvement of connectivity in hilly areas such as Himachal Pradesh, Uttarakhand, Kashmir, and parts of north-east will be very crucial. These reasons facilitate major off-season production.

- » **Special MSP for TOP Crops** - supported by government led production - This will ensure stability in acreage.
- » **Improved food processing** - Linking tomato value chain to processing of at least 10% of tomato production into tomato paste and puree during peak seasons, and using them in lean season (July-Aug), when tomato prices are high.
- » **Reforming Marketing:**
 - **Eliminating middleman and encouraging FPOs** to sell produce directly
 - **Amending APMC laws** to reduce cartelization, commission and other fees.
- » **Increasing Yield:**
 - Tomato yield in India at 25 tonnes per hectare (25t/ha) is very low than the global average of 37t/ha.
 - ICRIER suggests encouraging cultivation in structures called poly houses and greenhouses (as done in many European countries), which can control pest attacks.
- » **Accurate weather forecasts** should be made accessible to farmers. If there are sufficient advanced predictions of heavy rain or floods, farmers can prematurely harvest and save at least a part of their crop that can come into the market.
- » **Satellite mapping of sowing area and dissemination of this data for better supply side planning** - Both over-production and under-production creates multiple wastages and should be avoided.